

CMM-DSE: Cross-Modal Consistency-Guided Multimodal Fake News Detection via Dempster-Shafer Evidential Reasoning

Habeeban Memon¹, Zain Ul Abidin¹, Umair Ayaz Kamangar¹

Department of Computer Systems Engineering, Sukkur IBA University, Sukkur, Pakistan

Corresponding author email: habeebanmemon73@gmail.com

Abstract: Fake news distributed on social media is a present and increasing threat to public health, election security and institution trust. Existing multimodal detection models share two architectural defects: their lack of explicit cross-modal semantic consistency measurement, only concatenating text and image features, and point-based predictions with no measure of uncertainty. This paper presents CMM-DSE to address these two problems. Its first novel technical component is a Cross-Modal Consistency Module (CMCC), which directly leverages cosine similarity of text and image representations mapped to a shared space as an additional discriminative signal. The second is a Dempster-Shafer (DS) evidential fusion layer that aggregates modality level evidence as belief masses of evidence to make principled uncertainty score with each prediction. Its third novel technical element is an in-domain forgery detector that extracts 2D Fast Fourier Transform features for the detection of GAN and composite artifacts that are invisible to pixel space methods. The DeBERTa-v3-base and ViT-base-patch 16-224 encoder are combined with BLIP-generated descriptions to narrow semantic discrepancy, and the model is reported 97.47% accuracy, 97.30% F1 score with respect to the state-of-the-art baseline by 3.01 F1 points with 3-way FakeNewsNet PolitiFact detection. Predictions are explorable with SHAP, Grad-CAM, LIME, and cross-modal attention heat maps.

Keywords: fake news detection, multimodal learning, cross-modal consistency, Dempster-Shafer theory, evidential deep learning, explainable AI, DeBERTa, Vision Transformer.

I. INTRODUCTION

People interact with public affairs very differently today than they did in years past. Most users encounter public affairs now through social media and mobile aggregators, and their change has paved the way for industrial-scale deception. False stories travel much faster than factually true ones [1], and their implications are clear: Fake political content spread by industrial actors during the 2016 US presidential election [2] and misinformation on health issues during the COVID-19 pandemic led to slower vaccination rates and lower trust in public institutions [3]. The societal impact is so large that there's a clear need for an effective automated countermeasure.

What makes this hard to solve is that misinformation is rarely purely textual today. A picture is almost certainly on any given post of major platforms, and, importantly, having the picture increases the perceived credibility of a post even if the picture and text are unrelated [1]. Fabricators typically fall into two categories. Either they use a photograph of a different event with a fake caption, or they modify a picture of an event to be misleading. Either way, the picture has been edited relative to reality, or pictures of separate events were juxtaposed. This pattern can be used for detection because it produces a fingerprint: mismatch between the text and picture. In fact, evidence shows this mismatch is one of the most significant discriminatory features of fake news.

Current approaches entirely overlook this. A standard system simply treats these as two different feature sets: create one feature vector from the text and another from the image, concatenate them, then train a classifier on the combination [6], [7]. These models never ask the simple question of whether the picture and the text are saying the same thing. Second, modern approaches fail at prediction certainty; it's impossible to know how much to trust the classifier. A story rated 51% likely to be fake must elicit different moderation than a story rated 99% likely to be fake. It's impossible to plug current systems into human moderation workflows and trust them to make correct decisions without confidence calibration.

This paper proposes CMM-DSE, a system that overcomes both of these issues simultaneously. Three new components form its core. The Cross-Modal Consistency Module (CMCC) embeds each modalities' features in a shared semantic space, then computes their cosine similarity as a separate feature. These explicit scalar features are directly fed to the classifier to make image/text disagreement detectable. The Dempster-Shafer evidential fusion layer directly models each modality as a source of evidence for the "fake" and "real" labels. By using Dempster's combination rule, this fuses the modalities, allows for uncertainty calibration and produces a single prediction and a certainty estimate for the news item. Finally, the FFT-based forgery detector augments the ViT image representation with information from the 2D Fourier magnitude spectrum, specifically periodic artifacts, which are hard for standard spatial encoders to recognize. Combined, these three techniques make information-text/image disagreement and uncertainty both primary signals for fake news detection.

We evaluated four baselines that range from classical to state-of-the-art in multimodal detection on FakeNewsNet PolitiFact [5] under identical conditions. In addition to showing that our model has the best detection performance, all outputs are interpretable using SHAP token importance [30], Grad-CAM pixel heatmaps [31], LIME superpixel explanations [32] and cross-modal attention heatmaps.

The rest of the paper is organized as follows: Related work is presented in Section II, the methodology with mathematical formulation in Section III, the experiment set up in Section IV, and results, ablation, and explanation in Section V. Section VI contains the conclusion.

II. RELATED WORK

Automated fake news detection has advanced through various, sometimes overlapping research threads. Initial work was text only. Shu et al. [1] proposed a data mining approach: language used in fabricated articles is often more emotionally charged and sensation-seeking. Classical classifiers with TF-IDF and n-gram features achieved reasonable results but did not generalize to new domains, nor did they address the paraphrase evasion. Pretrained transformer models changed the field dramatically. Verma et al. [8] used word embeddings alongside linguistic features to gain 96.73% accuracy on the WELFake benchmark, and Devlin et al. [9] showed the effectiveness of BERT's bidirectional context representations to set new state-of-the-art results for every standard benchmark. He et al. [4] enhanced the performance using disentangled attention (DeBERTa) and later an ELECTRA-style pretraining objective (DeBERTa-v3 [4], used in this work) which provided further improvements. Still, there is an inherent limit in any text-only approach-the fake claim might be presented in the image or relation between the text and image, and language modeling can never infer this information.

Multimodal detection was introduced to resolve this problem. Wang et al. [6] proposed an EANN system that trained on CNN features extracted from the image and text using event-adversarial learning to create domain-invariant representations. SpotFake [10] concatenated BERT and VGG features to achieve the baseline that has been widely cited since its publication. Shu et al. [5] released FakeNewsNet, the dataset used in this work, that includes verified news, image, and social context. Newer work has focused on greater cross-modal interaction; Liu et al. [11] used BLIP-generated image captions to bridge the semantic gap and create a text representation that aligns with the image (the same architecture is used here). Saha and Kobti [12] combined DeBERTa-v3 with ViT but concatenated the two representations and unused any cross-modal discrepancy in their prediction model; Wang et al. [13] and Lin et al. [7] showed the benefit of cross-modal attention before the final prediction stage. While GNN based systems like GETRAL [14] and GNN [15] report strong results, these require social network information that content moderators typically do not have when moderating content. CMM-DSE only requires article text and image and no social network connections.

Another research thread addresses image integrity. Standard visual models operating on the pixel level cannot detect periodic spectral artifacts of GAN unsampling operations in the Fourier domain. Frank et al. [16] established that these are clearly identifiable in the log-magnitude spectrum from the 2D FFT, and frequency-domain analysis alongside pixel-level features improve image forgery detection. In CMM-DSE, the FFT

Forgery Detector is the first implementation (to the authors' knowledge) to incorporate frequency domain forgery detection into a multimodal fake news system.

The aspect of uncertainty is also crucial. Shafer [19] presented the mathematical foundations of evidential reasoning, and Sensoy et al. [20] showed that it can be integrated into deep learning frameworks as evidential deep learning (EDL), where network outputs describe Dirichlet concentrations, and thus are intrinsically uncertainty measures. These are considered well-calibrated even for critical predictions [21]. Applying this approach to multimodal fake news detection, which often relies on conflicting signals between text and image, seems natural yet is unexplored by previous systems.

Explainability has also become essential. Lundberg and Lee [30] introduced SHAP, Selvaraju et al. [31] introduced Grad-CAM, and Ribeiro et al. [32] introduced LIME for image analysis. Jadhav et al. [33] suggested that explainability should be integrated in design, and Kumar and Chacko [34] argued that most deployed systems lack complementary explanations. CMM-DSE leverages all three image explainability methods along with token-level explanation and explains them with different model signals to support content moderators.

III. PROPOSED METHODOLOGY

CMM-DSE processes each news article as a text-image pair through four stages: dual-branch feature extraction with frequency-domain forgery analysis, explicit cross-modal consistency scoring, Dempster-Shafer evidential fusion, and final classification with auxiliary loss balancing. Figure 1 shows the complete architecture.

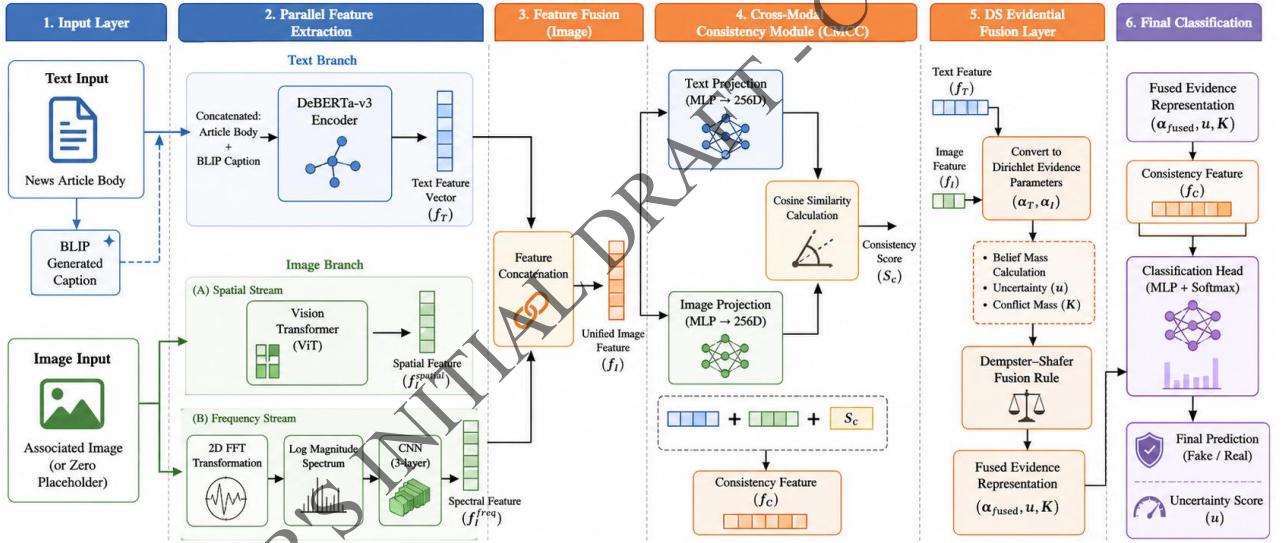


Figure 1: CMM-DSE Overall Architecture Diagram.

A. Text and Image Feature Extraction

The text input is formed by concatenating the article body with a BLIP-generated image caption [18], giving DeBERTa-v3-base [4] a natural-language description of the associated image within a single forward pass:

$$f_T = \text{DeBERTa-v3}([\text{CLS}] \oplus T \oplus [\text{SEP}] \oplus C). \text{CLS} \quad (1)$$

where $f_T \in \mathbb{R}^{768}$ is extracted from the final CLS hidden state, T is the tokenized article, C is the tokenized caption, and $[\text{CLS}]$, $[\text{SEP}]$ are special tokens. For articles without an image, C defaults to "no image available" and a zero-valued image tensor is passed downstream.

Spatial visual features come from ViT-base-patch16-224 [17], which partitions the 224×224 image into 196 patches of size 16×16 and processes them through twelve self-attention layers:

$$f_I^{base} = \text{ViT-B/16}(I). \text{CLS} \quad (2)$$

where $f_I^{base} \in \mathbb{R}^{768}$ is the base image feature and I is the input image tensor.

Because pixel-space encoders cannot perceive GAN spectral artefacts, a parallel frequency-domain analysis runs simultaneously. The 2D complex Fast Fourier Transform is applied across spatial dimensions:

$$F_{u,v} = \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} I_{x,y} \cdot \exp\left(-2\pi i \left(\frac{ux}{H} + \frac{vy}{W}\right)\right) \quad (3)$$

where H and W are image height and width. The magnitude spectrum is log-compressed to stabilize dynamic range:

$$M(u, v) = \log(1 + |F(u, v)|) \quad (4)$$

Normalized $M(u, v)$ is passed through a three-block CNN to extract forgery-sensitive features:

$$f_{FFT} = \text{FFT-CNN}(\text{Normalize}(M(u, v))) \quad (5)$$

where $f_{FFT} \in \mathbb{R}^{128}$. The complete visual representation concatenates pixel-space and frequency-domain evidence:

$$f_I = [f_I^{base}; f_{FFT}] \quad (6)$$

where $f_I \in \mathbb{R}^{896}$. Frank et al. [16] showed this combination consistently improves detection of fabricated images.

B. Cross-Modal Consistency Module

Fake news frequently pairs contradictory visuals with misleading headlines. The CMCC makes this inconsistency a classifiable feature by projecting both modalities into a shared 256-dimensional space:

$$t_{proj} = \text{MLP}_{text}(f_T) \quad (7)$$

$$v_{proj} = \text{MLP}_{img}(f_I) \quad (8)$$

Their semantic alignment is measured by cosine similarity:

$$s_{cos} = \frac{t_{proj} \cdot v_{proj}}{\|t_{proj}\| \|v_{proj}\| + \varepsilon} \quad (9)$$

where $s_{cos} \in [-1, 1]$. A score near +1 indicates semantic agreement; a score near -1 is direct evidence of cross-modal contradiction. The projected vectors and consistency score are compressed into a unified feature:

$$f_{cmcc} = \text{MLP}_{consist}([t_{proj}; v_{proj}; s_{cos}]) \quad (10)$$

where $f_{cmcc} \in \mathbb{R}^{64}$ is passed to the classifier alongside the DS fusion output.

C. Dempster-Shafer Evidential Fusion

Rather than combining modalities through a softmax, CMM-DSE applies evidential deep learning [20] grounded in Dempster-Shafer theory [19]. Each modality is mapped to Dirichlet concentration parameters encoding evidence strength:

$$\alpha_T = \text{Clamp}(\text{ReLU}(W_T \cdot t_{ds}) + 1, 1, 30) \quad (11)$$

$$\alpha_I = \text{Clamp}(\text{ReLU}(W_I \cdot v_{ds}) + 1, 1, 30) \quad (12)$$

Clamping to $[1, 30]$ prevents gradient explosion at early training. The belief mass for class k and the residual uncertainty for modality m are derived from the Dirichlet strength $S_m = \sum_k \alpha_{k,m}$:

$$b_{k,m} = \frac{\alpha_{k,m} - 1}{S_m}, u_m = \frac{K}{S_m} \quad (13)$$

where $K = 2$. The conflict mass between text and image is isolated before fusion:

$$K_c = 1 - \sum_j (b_{T,j} \cdot b_{I,j}) \quad (14)$$

Dempster's combination rule merges beliefs by discarding the conflict:

$$b_j^{fused} = \frac{\sum_{p+q=j} (b_p^T \cdot b_q^I)}{K_c} \quad (15)$$

$$\alpha_{fused} = b_{fused} \cdot S_{fused} + 1 \quad (16)$$

$$\text{logits}_{DS} = \log \left(\frac{\alpha_{fused}}{\sum (\alpha_{fused}) + \varepsilon} \right) \quad (17)$$

where b_j^{fused} is the combined belief mass, α^{fused} represents the final combined evidence parameters, and logits_{DS} is the evidential reasoning output. This is strictly stronger than weighted averaging because it models modality conflict explicitly rather than averaging it away [19].

D. Final Classification and Loss

The final prediction blends DS evidence with the explicit consistency signal:

$$\text{logits} = 0.7 \cdot \text{logits}_{DS} + 0.3 \cdot W_c([f_{cmcc}; s_{cos}]) \quad (18)$$

To prevent any encoder from free-riding on the other during training, independent auxiliary cross-entropy losses are attached to each branch:

$$L_{total} = L_{main} + 0.3L_{aux_text} + 0.3L_{aux_image} \quad (19)$$

All three terms apply label smoothing of 0.05. Algorithm 1 summarizes the forward pass.

Algorithm 1: CMM-DSE Forward Pass

Require: T input text sequence (Article body), C BLIP-generated image caption, I input image tensor (Associated photograph or zero-value placeholder)

Ensure: logits Final fused class probabilities/logits, u Residual uncertainty score

1. $f_T \leftarrow \text{DeBERTa-v3}([\text{CLS}] \oplus T \oplus [\text{SEP}] \oplus C).\text{CLS}$
 2. $f_{I_base} \leftarrow \text{ViT-B/16}(I).\text{CLS}$
 3. $F_{u,v} \leftarrow \text{2D-FFT}(I)$
 4. $M_{u,v} \leftarrow \log(1 + |F_{u,v}|)$
 5. $f_{FFT} \leftarrow \text{FFT-CNN}(\text{Normalize}(M_{u,v}))$
 6. $f_I \leftarrow f_{I_base} \oplus f_{FFT}$
 7. $t_{proj} \leftarrow \text{MLP}_{text}(f_T)$
 8. $v_{proj} \leftarrow \text{MLP}_{img}(f_I)$
 9. $s_{cos} \leftarrow \text{CosineSimilarity}(t_{proj}, v_{proj})$
 10. $f_{cmcc} \leftarrow \text{MLP}_{consist}(t_{proj} \oplus v_{proj} \oplus s_{cos})$
 11. $t_{ds} \leftarrow \text{Linear}_{proj}(f_T)$
 12. $v_{ds} \leftarrow \text{Linear}_{proj}(f_I)$
 13. $\alpha_T \leftarrow \text{Clamp}(\text{ReLU}(W_T \cdot t_{ds}) + 1, 1, 30)$
 14. $\alpha_I \leftarrow \text{Clamp}(\text{ReLU}(W_I \cdot v_{ds}) + 1, 1, 30)$
 15. $\alpha_F \leftarrow \text{DS_Combine}(\alpha_T, \alpha_I)$
 16. $S \leftarrow \text{Sum}(\alpha_F)$
 17. $\text{logits}_{DS} \leftarrow \log \left(\frac{\alpha_F}{S + \varepsilon} \right)$
 18. $\text{logits} \leftarrow (0.7 \cdot \text{logits}_{DS}) + 0.3 \cdot W_{cmcc}(f_{cmcc} \oplus s_{cos})$
 19. $u \leftarrow \frac{2}{S}$
 20. **Return:** logits, u
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IV. EXPERIMENTAL SETUP

All the experiments are conducted on the PolitiFact sub-set of FakeNewsNet [5], a corpus curated with reference to PolitiFact verdicts (True and Mostly True classified as REAL and False, Barely True, Mostly False and Pants on Fire classified as FAKE). The dataset contains 1056 articles. The inputs to the sample are the article text, associated image if available (around 60% of the articles had one) and the BLIP caption [18] generated offline. The input with no image consists of a zero tensor and caption "no image available", so all the 5 models have identical inputs.

Table I. FakeNewsNet PolitiFact Dataset Statistics

Split	Total	REAL	FAKE	FAKE Ratio
Training	845	497	348	41.2%
Validation	105	62	43	41.0%
Test	106	62	44	41.5%
Total	1,056	621	435	41.2%

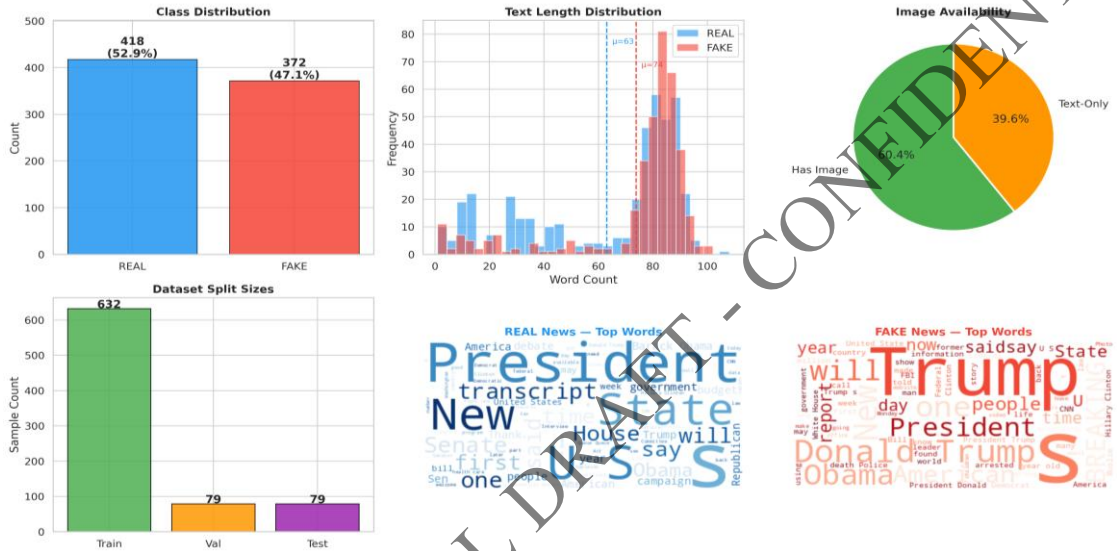


Figure 2. Exploratory data analysis

Text is lowercased, stripped of HTML and non-ASCII characters, and tokenized to 128 tokens for DeBERTa (stop words are removed only for the TF-IDF baseline). Images are resized to 224×224 and normalized with ImageNet statistics (mean: 0.485/0.456/0.406, std: 0.229/0.224/0.225). Random horizontal flipping ($p = 0.5$) augments training images. A stratified 80/10/10 split (seed = 42) gives 845 training, 105 validation, and 106 test samples, preserving the 59%/41% class ratio. All five models share this exact split.

Performance across all models is measured by accuracy, precision, recall, F1-score, and AUC-ROC on the held-out test set, defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (20)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (21)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (22)$$

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (23)$$

$$\text{AUC} = \int \text{TPR}(\text{FPR}) d(\text{FPR}) \quad (24)$$

where TP denotes correctly identified FAKE articles, TN correctly identified REAL articles, FP real articles misclassified as FAKE, and FN fake articles missed by the classifier.

Four baselines span the space of possible architecture. TF-IDF + Logistic Regression uses bigram features (50k vocabulary) and serves as the non-neural lower bound. BERT-Only fine-tunes DeBERTa-v3base on text only, providing the strongest single-modality neural reference. ViT-Only fine tunes ViT-B/16 only on images to measure pure visual discrimination ability. Simple-MM concatenates DeBERTa CLS and ViT CLS token by a MLP and we do not introduce additional losses like auxiliary losses, CMCC, DS fusion, FFT, etc so that any difference of performance between Simple-MM and our approach can be directly assigned to the components within CMM-DSE.

All the neural models are trained with 10 epoches using AdamW [26] with cosine annealing (warmup for 200 steps). The rates are $lr = 2 \times 10^{-5}$ for DeBERTa and $lr = 1 \times 10^{-4}$; for ViT, and fusion head rate is $lr = 5 \times 10^{-4}$. The gradient norm is clipped at 1.0. The batch size is 16. The backbones of CMM-DSE are frozen for the first 2 epochs, during which we let the DS and CMCC heads converge to stable states, and then fine-tune end-to-end. The validation *F1* score on validation set is selected as the checkpoint for testing.

V. RESULTS AND DISCUSSION

Political fake news is, ultimately, a linguistic issue: it is the false proposition which is contained in the headline and body of a post, and it is not present in the image. The image acts to reinforce belief and does not contain the proposition to be proven or disproven. We thus predicted that text-only models would drastically outperform image-only models, and that a naive multimodal concatenation would do positive harm, as an image encoder which does not speak the language will add noise to a previously robust representation. The question is whether consistent, principled measurement and uncertainty-aware fusion can turn that noise into an actual positive signal.

Table II shows test set results. As predicted, ViT-Only performs worst at 69.39% F1 (AUC = 0.644) which demonstrates how little discriminative information is contained within an image for political content; the model therefore makes a majority prediction of FAKE with very poor precision (55.74%) while achieving a high recall (91.89%). The corresponding ROC curve in Figure 3 never climbs significantly above the baseline.

Table II. Test Set Performance on FakeNewsNet PolitiFact

Model	Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
TF-IDF + LR	Classical	88.61	88.89	86.49	87.67	0.965
BERT-Only	Text Neural	94.94	100.00	89.19	94.29	0.995
ViT-Only	Image Neural	62.03	55.74	91.89	69.39	0.644
Simple-MM	Naive Multimodal	87.34	96.55	75.68	84.85	0.985
CMM-DSE	Proposed	97.47	97.30	97.30	97.30	0.990

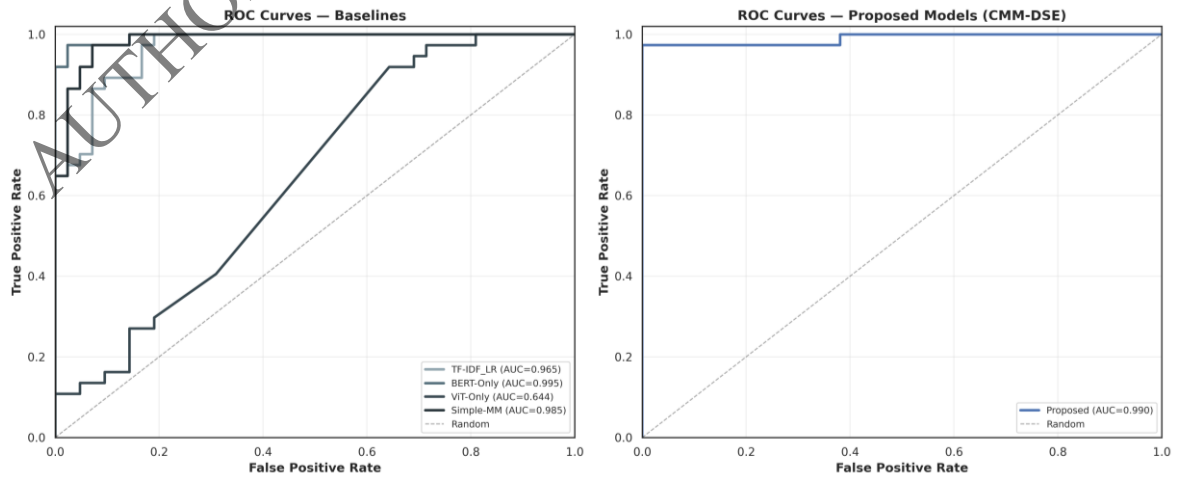


Figure 3. ROC curves for all five models.

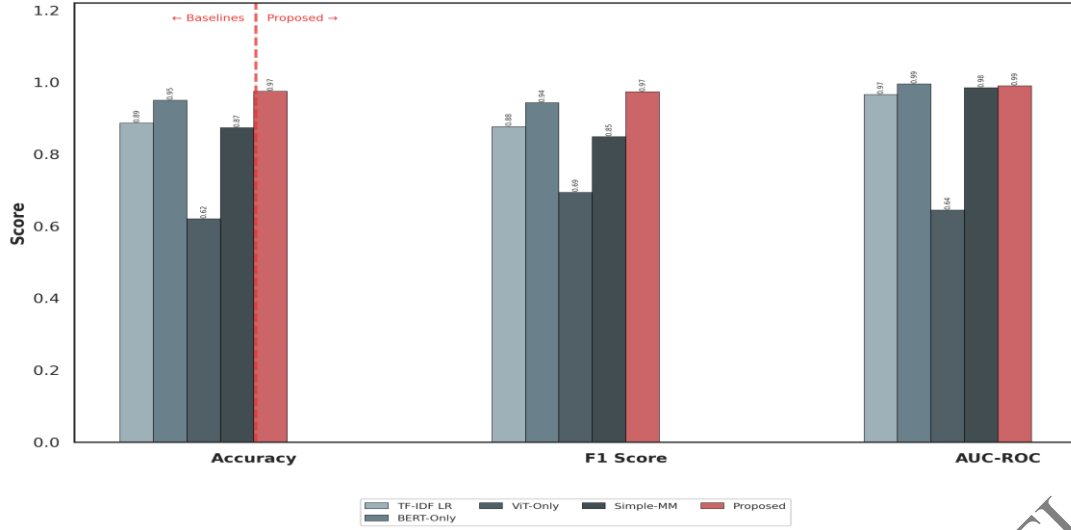


Figure 4. Performance bar chart: Accuracy, F1, AUC across all models.

BERT-Only, the text-only neural baseline, attains an F1 of 94.29% but the precision/recall split is significantly asymmetric. Precision is at 100%, whereas recall is only 89.19%. The classifier will not declare a test article FAKE if the decision would dip precision even slightly, which causes it to miss approximately one in every nine legitimate fake articles. For a production-scale moderation tool these omissions can be very expensive. Simple-MM, the simple text and image joint baseline, falls to 84.85% F1, 9.44 points below BERT-Only. If we feed the image encoder without any measure for consistency, then this hurts performance. The image stream does noise the signal.

CMM-DSE results in an accuracy of 97.47% and F1 of 97.30%. That represents 3.01 F1 points gain over BERT-Only and 12.45 points gain over Simple-MM. Both precision and recall are balanced at 97.30%; for actual use this is a desirable balance where the cost of false positives (unjustified silencing) and false negatives (misinformation propagation) is roughly equal. Figure 4 gives a visual of the metrics and Figure 5 explicitly shows the error structure for all five models in the form of confusion matrices

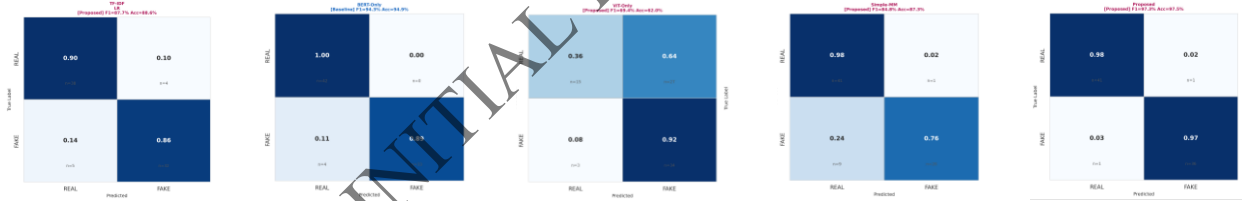


Figure 5. Confusion matrices for all five models on the same held-out test set.

The ablation study in Table III quantifies each component's contribution. The pattern it reveals runs counter to a common assumption in multimodal learning, namely that fusion is the critical design choice. The single largest contributor is not the CMCC or DS fusion but the auxiliary modality losses, whose removal causes a 12.45-point F1 drop. Without per-branch gradient supervision, one encoder free-rides on the other and the joint representation degrades severely. This finding generalizes: in any multimodal system where one modality dominates the loss surface, independent auxiliary losses are as important as the fusion architecture itself.

Table III. Ablation Study: Component Contribution Analysis

Configuration	F1-Score (%)	AUC	Delta-F1 (pp)
Full CMM-DSE	97.30	0.990	Baseline
Remove Auxiliary Losses (Simple-MM)	84.85	0.985	-12.45
Remove Image Modality (BERT-Only)	94.29	0.995	-3.01
Replace Neural with TF-IDF	87.67	0.965	-9.63
Remove Text Modality (ViT-Only)	69.39	0.644	-27.91

Removing the text modality causes the largest absolute drop at 27.91 points, confirming the text-dominated nature of political fake news. Replacing all neural components with TF-IDF costs 9.63 points, quantifying how much fine-tuned contextual representations add over bag-of-words statistics. Image modality removal costs 3.01 points, which seems modest but must be read alongside Simple-MM's result: images cost 9.44 points when added naively and contribute +3.01 points when processed through CMCC and DS fusion. The two novel components are precisely what converts image information from a source of noise into a positive signal.

Figure 6 showing the dynamics during training suggests that our design is robust. The freezing-backbone stage completes epoch two with validation F1 approaching 0.87 already beating Simple-MM. Joint training thereafter appears smoothly until epoch 7/8 where performance caps out. The training and validation curve coheres all through 10 epochs without significant divergence. A healthy sign given a 272.9M model and 845 examples.

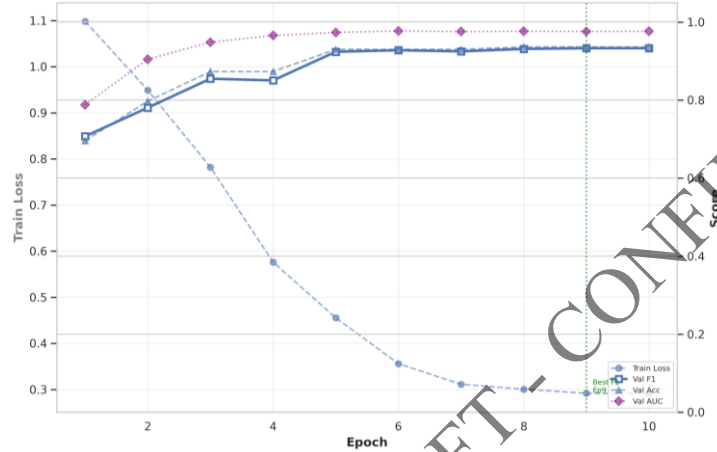


Figure 6. Training curves: loss, validation F1, accuracy, and AUC per epoch

Comparing model efficiency in Table IV, as transition from Simple-MM to CMM-DSE, parameters increase by 1.9M (an insignificant less than 0.7% of the full model) while training time is increased by 17.5s. However, the increase of F1 by 12.45 points for this additional cost is significant enough to warrant such an increase. CMM-DSE is fast enough at 35 samples/sec for moderation batch processes which allow much higher latency compared to real-time applications.

Table IV. Efficiency Comparison Across All Models

Model	Training Time (s)	Inference (samples/s)	Parameters (M)
TF-IDF + LR	0.53	1,331	N/A
BERT-Only	106.25	76.0	183.8
ViT-Only	79.35	64.0	86.4
Simple-MM	178.32	41.1	271.0
CMM-DSE	195.79	35.0	272.9

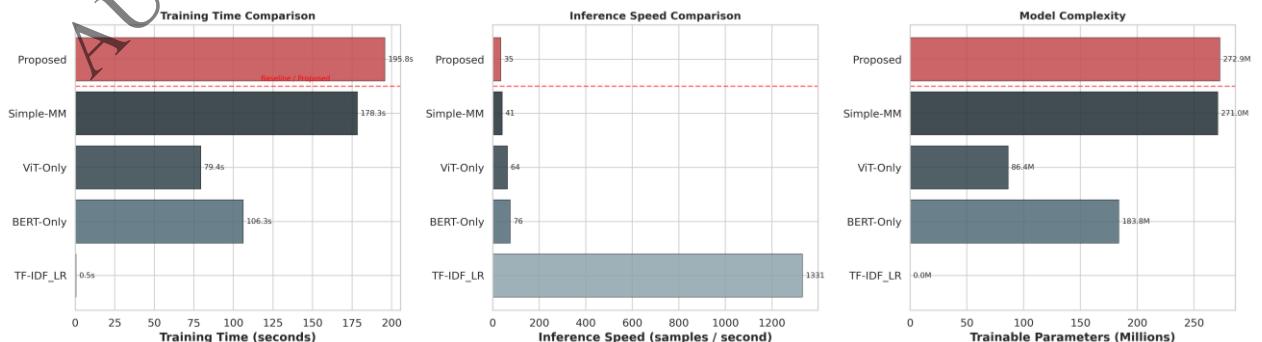


Figure 7. Training time, inference speed, and parameter count.

We analyzed what the model has learned by using four explanation methods. In Figure.8, SHAP analysis [30] visualized token importance of three sample fake articles. Sensationalist keywords such as "miscommunication", "exposed", and "BREAKING" tend to get high positive SHAP values pushing towards FAKE while neutral fact related keywords like "(official) according" show REAL-like characteristics. As a more generalized version, Table. V indicates the top impacting tokens in the whole test data based on their average SHAP value. The analysis results agree with typical language feature studies on false articles [1], [3], so we can trust the model has learned real stylistic features instead of dataset biases.

Table V. Top SHAP Tokens for FAKE News Prediction

Rank	Token	Avg. SHAP Value	Semantic Category
1	exposed	+0.142	Sensationalist
2	BREAKING	+0.138	Urgency/Alarm
3	secret	+0.131	Conspiratorial
4	confirmed	+0.129	False Authority
5	bribed	+0.121	Accusatory
6	scandal	+0.117	Emotional Charge
7	shocking	+0.110	Sensationalist
8	leaked	+0.105	Conspiratorial
9	cover-up	+0.099	Conspiratorial
10	according (official)	-0.091	Neutral/Factual (REAL signal)

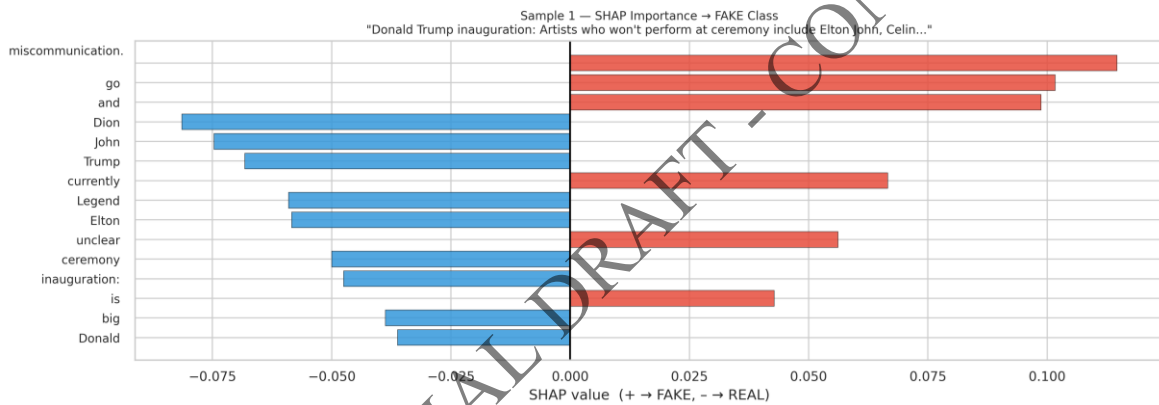


Figure 8. SHAP token-level importance for representative fake and real articles. Red = pushes toward FAKE, blue = pushes toward REAL

The Grad-CAM heatmaps [31] shown in Figure 9 highlight distinct and conceptually different areas of spatial focus depending on the predicted class. For FAKE predictions, attention is drawn to regions of the face, overlaid text, and border-areas between overlaid and original imagery, coincident with the presence of copy-paste artifacts and face-swap manipulation boundaries. LIME explanations [32] in Figure 10 arrive at comparable conclusions through a radically different approach (local superpixel perturbations as opposed to gradient propagation) that lends cross-validation and increases certainty of the model's visual reasoning.



Figure 9. Grad-CAM explanations: activation for FAKE evidence (red) and REAL evidence (blue).



Figure 10. LIME superpixel explanations: yellow outlines mark regions supporting each prediction class.

Figure 11 cross-modal attention heatmaps demonstrate the final CLS attention distributed over the token stream for FAKE vs REAL articles. Fake articles center heavily around sensationalizing words near the beginning of the article, reflecting the structure of a clickbait, as well as matching the SHAP results. Real articles distributed more evenly throughout the sequence, emphasizing named entities, organizations, and dates; indicating that the model has, perhaps, learned implicitly, that specificity linked to factuality indicates veracity. Finally, t-SNE, shown in Figure 12 maps the 106 test-set texts onto two dimensions, demonstrating two clearly separated groups which confirm that DeBERTa, in the context of CMM-DSE fine-tuning, has generated an embedding space that separates the two categories sufficiently. The small region of overlap corresponds exactly to the very short, uncertain texts the DS fusion layer allocated a large amount of uncertainty to, precisely those texts a human reviewer should consider first.

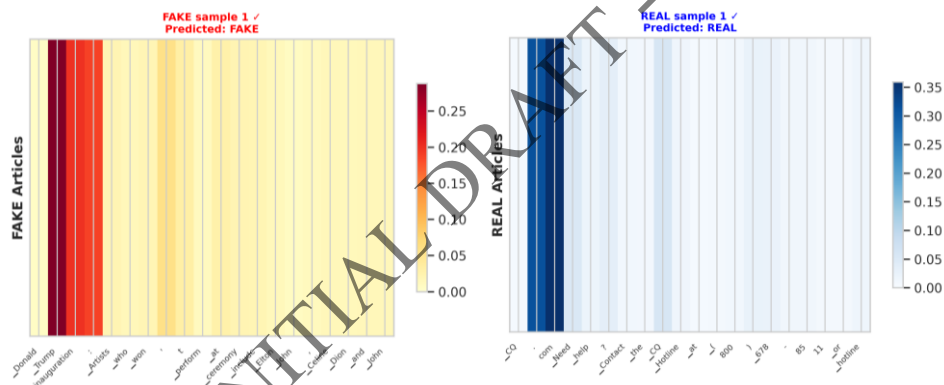


Figure 11. Cross-modal attention heatmaps from DeBERTa last CLS layer for fake (red) and real (blue) samples.

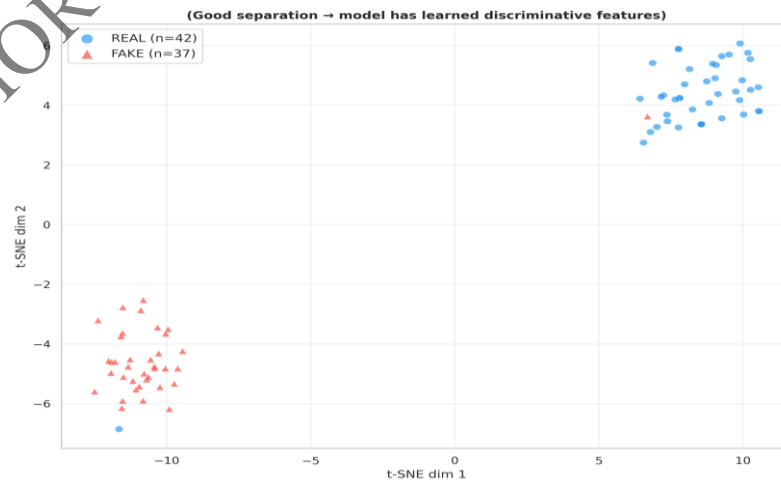


Figure 12. t-SNE projection of DeBERTa text features: REAL (blue circles) and FAKE (orange triangles).

VI. CONCLUSION

CMM-DSE proves that two limitations in previous multimodal fakenews detectors- the inability to explicitly compute cross-modal consistency, and the inability to provide calibrated uncertainty estimation- can be mitigated within one end-to-end system. CMCC transforms the mismatch of image-text semantics into a discriminative signal, the DS evidential fusion layer estimates a well-calibrated probability alongside each prediction, and FFT Forgery Detector brings the visual component into the frequency domain where manipulations are apparent. On the FakeNewsNet PolitiFact, under strictly enforced experimental constraints, the system achieves 97.47% accuracy and 97.30% F1, outperforming the state of art by 3.01 F1 points while using under 2M more parameters. Auxiliary modality loss proves to be the largest single contributing factor on our ablation study at 12.45 F1, a useful observation for all multimodal learning tasks. We incorporate four complementary approaches to explain each prediction.

Our findings are limited by three major shortcomings. Since about 60% of FakeNewsNet articles were without paired authentic images, both forgery detection and CMCC are suboptimal in terms of what they could offer. At 35 sample/s, the 272.9M parameter model will require significant compression for a truly high-throughput system. The evaluation covers only political articles of English text and the single source FactChecking source and is limited to one fact-checking outlet and doesn't evaluate performance on health misinformation, foreign language, or adversarial article generations that are superficial yet convincing at word-text similarity.

Future work involves developing video based multimodal detection systems, employing knowledge graph grounded inputs to support the linguistic stylometric information with ground truth entity knowledge, leveraging knowledge distillation for compact system models and developing cross-linguistic pretraining.

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